

```
df_doctored <- df %>%
  filter((ticker != "NFLX" | call_put != "calls") | (K < 266 & true_price < 400)) %>%
  filter((ticker != "NVDA" | call_put != "calls") | (K < 500 & true_price < 300)) %>%
  filter((ticker != "NFLX" | call_put != "puts") | (K < 266 & (K < 100 | true_price > 10))) %>%
  filter((ticker != "NVDA" | call_put != "puts") | (K < 500 & (K < 300 | true_price > 100)))

df <- df_doctored
```

Regresní analýza

Absolutní

```
df_absolute <- df_doctored %>%
  pivot_longer(cols = c(user_bs, user_crr, user_mc),
    names_to = "model",
    values_to = "model_price") %>%
  #filter(model == "user_crr") %>%
  mutate(pricing_error = true_price - model_price) %>%
  filter(is.finite(pricing_error)) %>%
  #filter(abs(pricing_error) <= 13) %>%
  mutate(snapshot_date = as.Date(expiration) - round(T * 365)) %>%
  mutate(snapshot_month = format(snapshot_date, "%m"))

df_absolute_temp <- df_absolute %>% filter(am_eu == "american" & call_put == "calls")
train_indexes <- sample(1:nrow(df_absolute_temp), size = 0.8 * nrow(df_absolute_temp))

train_data_abs <- df_absolute_temp[train_indexes, ]
test_data_abs <- df_absolute_temp[-train_indexes, ]
```

Prostý model

```
subset_args <- c(
  "S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  #"am_eu",
  "call_put",
  #"model",
  "ticker",
  "snapshot_month"
)
interactions <- c(
  #"q:am_eu",
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))
#T_crr <- lm(pricing_error ~ T + as.factor(snapshot_month), data = train_data)
```

```
regression_absolute <- lm(formula, data = train_data_abs)

summary(regression_absolute)
```

```
##
## Call:
## lm(formula = formula, data = train_data_abs)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -926.59   -2.00     1.61     5.69   994.81
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.493e+01  3.774e+00   6.607 3.93e-11 ***
## S              -3.750e-02  1.185e-03  -31.639 < 2e-16 ***
## moneyness       7.988e-02  6.829e-03   11.697 < 2e-16 ***
## K               1.386e-02  2.581e-04   53.685 < 2e-16 ***
## r               8.779e+02  5.542e+01   15.842 < 2e-16 ***
## sigma          -5.146e+00  2.172e+00   -2.369  0.0178 *
## q              -7.074e+02  9.861e+01   -7.173 7.33e-13 ***
## T               1.506e+00  8.062e-02   18.681 < 2e-16 ***
## call_putputs    3.113e+00  1.138e-01   27.342 < 2e-16 ***
## ticker^VIX      -5.444e+01  4.351e+00  -12.513 < 2e-16 ***
## ticker^XDA      -5.793e+01  4.974e+00  -11.648 < 2e-16 ***
## ticker^XDB      -5.620e+01  4.310e+00  -13.040 < 2e-16 ***
## ticker^XDE      -5.646e+01  3.735e+00  -15.117 < 2e-16 ***
## ticker^XDN      -5.668e+01  4.088e+00  -13.863 < 2e-16 ***
## tickerAAPL      -5.024e+01  2.904e+00  -17.303 < 2e-16 ***
## tickerAMD        -5.212e+01  3.138e+00  -16.609 < 2e-16 ***
## tickerAMZN       -5.284e+01  2.975e+00  -17.761 < 2e-16 ***
## tickerGOOGL     -4.486e+01  2.847e+00  -15.760 < 2e-16 ***
## tickerJNJ        -3.045e+01  4.270e+00   -7.132 9.89e-13 ***
## tickerKO         -3.612e+01  4.150e+00   -8.704 < 2e-16 ***
## tickerMETA       -4.519e+01  2.543e+00  -17.774 < 2e-16 ***
## tickerMSFT       -4.560e+01  2.772e+00  -16.454 < 2e-16 ***
## tickerNFLX       -5.807e+01  3.133e+00  -18.532 < 2e-16 ***
## tickerNVDA       -5.270e+01  3.071e+00  -17.157 < 2e-16 ***
## tickerPG         -3.620e+01  3.894e+00   -9.297 < 2e-16 ***
## tickerTSLA       -5.213e+01  2.939e+00  -17.737 < 2e-16 ***
## snapshot_month02  1.493e-01  1.654e-01    0.903  0.3667
## snapshot_month03 -1.537e-01  1.843e-01   -0.834  0.4042
## snapshot_month05  2.765e+00  2.244e-01   12.321 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.28 on 375333 degrees of freedom
## Multiple R-squared:  0.01882,    Adjusted R-squared:  0.01875
## F-statistic: 257.2 on 28 and 375333 DF,  p-value: < 2.2e-16
```

Předpoklady

Kontrola normality dat:

```
png("~/Praxe/bakalarka/figures/hist_absolute.png", width = 6, height = 4, units = "in", res = 300)
histogram_abs <- hist(df_absolute_temp$pricing_error, breaks = 100)
dev.off()
```

```
## png
## 2
```

```
histogram_abs
```

```
## $breaks
## [1] -960 -940 -920 -900 -880 -860 -840 -820 -800 -780 -760 -740 -720 -700 -680
## [16] -660 -640 -620 -600 -580 -560 -540 -520 -500 -480 -460 -440 -420 -400 -380
## [31] -360 -340 -320 -300 -280 -260 -240 -220 -200 -180 -160 -140 -120 -100 -80
## [46] -60 -40 -20 0 20 40 60 80 100 120 140 160 180 200 220
## [61] 240 260 280 300 320 340 360 380 400 420 440 460 480 500 520
## [76] 540 560 580 600 620 640 660 680 700 720 740 760 780 800 820
## [91] 840 860 880 900 920 940 960 980 1000
##
## $counts
## [1] 6 0 0 3 0 0 3 0 0 6
## [11] 15 3 21 12 3 27 13 29 15 27
## [21] 51 39 33 18 36 33 58 65 61 112
## [31] 112 93 96 120 143 240 197 268 322 555
## [41] 441 824 1375 1497 2823 5189 13693 118267 300957 12942
## [51] 3972 1879 956 567 297 116 127 95 21 24
## [61] 39 0 17 5 17 21 18 0 6 3
## [71] 6 0 6 15 3 3 9 24 3 19
## [81] 5 0 6 9 7 12 12 8 8 7
## [91] 0 0 0 0 3 12 0 3
##
## $density
## [1] 6.393821e-07 0.000000e+00 0.000000e+00 3.196911e-07 0.000000e+00
## [6] 0.000000e+00 3.196911e-07 0.000000e+00 0.000000e+00 6.393821e-07
## [11] 1.598455e-06 3.196911e-07 2.237837e-06 1.278764e-06 3.196911e-07
## [16] 2.877219e-06 1.385328e-06 3.090347e-06 1.598455e-06 2.877219e-06
## [21] 5.434748e-06 4.155984e-06 3.516602e-06 1.918146e-06 3.836293e-06
## [26] 3.516602e-06 6.180694e-06 6.926639e-06 6.500385e-06 1.193513e-05
## [31] 1.193513e-05 9.910423e-06 1.023011e-05 1.278764e-05 1.523861e-05
## [36] 2.557528e-05 2.099305e-05 2.855907e-05 3.431351e-05 5.914284e-05
## [41] 4.699458e-05 8.780848e-05 1.465251e-04 1.595258e-04 3.008293e-04
## [46] 5.529590e-04 1.459177e-03 1.260297e-02 3.207109e-02 1.379147e-03
## [51] 4.232710e-04 2.002332e-04 1.018749e-04 6.042161e-05 3.164941e-05
## [56] 1.236139e-05 1.353359e-05 1.012355e-05 2.237837e-06 2.557528e-06
## [61] 4.155984e-06 0.000000e+00 1.811583e-06 5.328184e-07 1.811583e-06
## [66] 2.237837e-06 1.918146e-06 0.000000e+00 6.393821e-07 3.196911e-07
## [71] 6.393821e-07 0.000000e+00 6.393821e-07 1.598455e-06 3.196911e-07
## [76] 3.196911e-07 9.590732e-07 2.557528e-06 3.196911e-07 2.024710e-06
## [81] 5.328184e-07 0.000000e+00 6.393821e-07 9.590732e-07 7.459458e-07
## [86] 1.278764e-06 1.278764e-06 8.525095e-07 8.525095e-07 7.459458e-07
## [91] 0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 3.196911e-07
## [96] 1.278764e-06 0.000000e+00 3.196911e-07
##
## $mids
## [1] -950 -930 -910 -890 -870 -850 -830 -810 -790 -770 -750 -730 -710 -690 -670
## [16] -650 -630 -610 -590 -570 -550 -530 -510 -490 -470 -450 -430 -410 -390 -370
```

```
## [31] -350 -330 -310 -290 -270 -250 -230 -210 -190 -170 -150 -130 -110 -90 -70
## [46] -50 -30 -10 10 30 50 70 90 110 130 150 170 190 210 230
## [61] 250 270 290 310 330 350 370 390 410 430 450 470 490 510 530
## [76] 550 570 590 610 630 650 670 690 710 730 750 770 790 810 830
## [91] 850 870 890 910 930 950 970 990
##
## $xname
## [1] "df_absolute_temp$pricing_error"
##
## $equidist
## [1] TRUE
##
## attr("class")
## [1] "histogram"
```

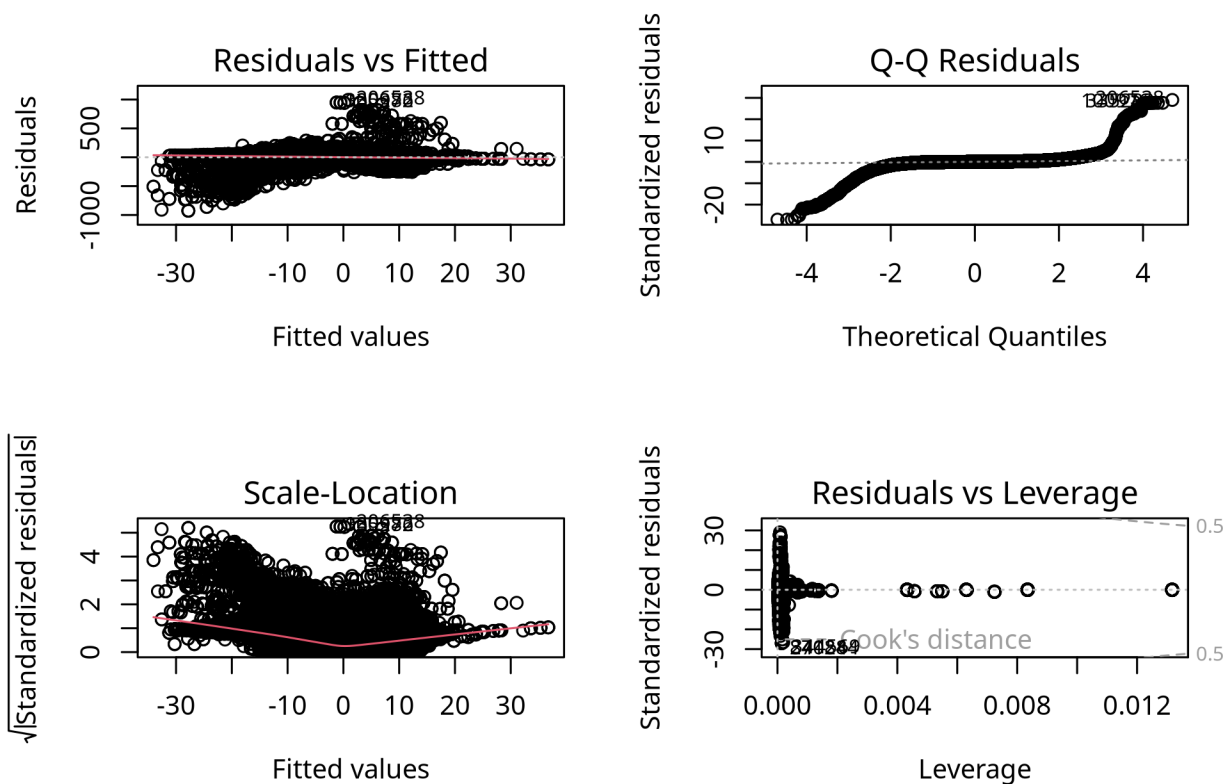
Data jsou normálně rozložená s velmi silnými ohony a slabě negativním sklonem.

Celková kontrola regresního modelu:

```
png("~/Praxe/bakalarka/figures/basic_lm_plots.png", width = 6, height = 4, units = "in", res = 300)
par(mfrow = c(2, 2))
plot(regression_absolute)
dev.off()
```

```
## png
## 2
```

```
par(mfrow = c(2, 2))
plot(regression_absolute)
```



```
par(mfrow = c(1, 1))
```

První graf ukazuje lineární funkci, ne konstantní funkci. Co to přesně znamená? QQ-plot (graf 2) potvrzuje přiměřeně vhodné rozdělení dat. Homoskedasticita silně porušena (graf 3). Cookovy vzdálenosti (graf 4) jsou ok.

Druhá kontrola homoscedasticity:

```
bptest(regression_absolute)
```

```
##
## studentized Breusch-Pagan test
##
## data: regression_absolute
## BP = 12434, df = 28, p-value < 2.2e-16
```

Dle výsledků Breuschova-Paganova testu ($p < 0.05$) zamítáme null hypotézu → je přítomná heteroskedasticita.

Použijeme robustní standardní chyby (HC3), které opravují heteroskedasticitu bez změny koeficientů:

Kontrola autokorelace:

```
dwtest(regression_absolute)
```

```
##
## Durbin-Watson test
##
## data: regression_absolute
## DW = 1.9917, p-value = 0.0054
## alternative hypothesis: true autocorrelation is greater than 0
```

Dle výsledku Durbinova-Watsonova testu je přítomná autokorelace. Opce ze stejného tickeru jsou navzájem závislé.

Použijeme cluster-robust standardní chyby (cluster na ticker), které opravují heteroskedasticitu i autokorelací zároveň:

Koeficienty zůstávají stejné jako v OLS, ale standardní chyby jsou opravené pro oba problémy.

Kontrola multikolinearity:

```
vif(regression_absolute, type = "predictor")
```

```
## GVIFs computed for predictors
```

```
##              GVIF Df GVIF^(1/(2*Df)) Interacts With
## S              2.573115e+02  1          16.040933      --
## moneyness      1.054855e+00  1           1.027061      --
## K              1.170557e+01  1           3.421340      --
## r              1.614553e+00  1           1.270651      --
## sigma          6.300589e+01  1           7.937625      --
## q              1.638284e+02  1          12.799549      --
## T              1.084767e+00  1           1.041521      --
## call_put       1.031814e+00  1           1.015782      --
## ticker         1.795057e+06 17           1.527367      --
## snapshot_month 4.241099e+00  3           1.272271      --
```

```
##              Other Predictors
## S              moneyness, K, r, sigma, q, T, call_put, ticker, snapshot_month
## moneyness      S, K, r, sigma, q, T, call_put, ticker, snapshot_month
## K              S, moneyness, r, sigma, q, T, call_put, ticker, snapshot_month
## r              S, moneyness, K, sigma, q, T, call_put, ticker, snapshot_month
## sigma          S, moneyness, K, r, q, T, call_put, ticker, snapshot_month
## q              S, moneyness, K, r, sigma, T, call_put, ticker, snapshot_month
## T              S, moneyness, K, r, sigma, q, call_put, ticker, snapshot_month
## call_put       S, moneyness, K, r, sigma, q, T, ticker, snapshot_month
## ticker         S, moneyness, K, r, sigma, q, T, call_put, snapshot_month
## snapshot_month S, moneyness, K, r, sigma, q, T, call_put, ticker
```

Extrémní VIF výsledek pro am_eu a call_put. Silná multikolinearita. Každý ticker má pouze jeden typ opce, am_eu je s ticker redundantní.

LMER model - základní

```
subset_args <- c(
  "S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model",
  "ticker"
)
interactions <- c(
  "q:am_eu",
```

```

  #"(1 | option_id)",
  "(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_absolute <- lmer(formula, data = train_data_abs)

## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
Náhodný efekt (1 | ticker) již částečně absorbuje autokorelací mezi opcemi stejného podkladového aktiva.
summary(rmr_absolute)

## Linear mixed model fit by REML ['lmerMod']
## Formula: pricing_error ~ S + moneyness + K + r + sigma + q + T + am_eu +
##      call_put + model + ticker + q:am_eu + (1 | ticker) + (1 |
##      snapshot_month)
##      Data: train_data_abs
##
## REML criterion at convergence: 3718786
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -27.0279  -0.0582   0.0468   0.1658  29.0164
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
##      ticker              (Intercept)      19.275   4.390
##      snapshot_month      (Intercept)       1.885   1.373
##      Residual                        1175.329  34.283
## Number of obs: 375362, groups:  ticker, 18; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept) -2.654e+01  5.040e+00  -5.266
## S            -3.736e-02  1.182e-03 -31.609
## moneyness     7.990e-02  6.829e-03  11.700
## K             1.386e-02  2.581e-04  53.689
## r             8.818e+02  5.531e+01  15.943
## sigma        -5.457e+00  2.165e+00  -2.521
## q            -7.071e+02  9.861e+01  -7.171
## T             1.506e+00  8.062e-02  18.686
## am_eueuropean 5.171e+01  6.865e+00   7.533
## call_putputs  3.113e+00  1.138e-01  27.341
## modeluser_crr -2.669e-02  1.371e-01  -0.195
## modeluser_mc  4.318e-02  1.370e-01   0.315
## ticker~VIX    -5.373e+01  7.570e+00  -7.098
## ticker~XDA    -5.760e+01  7.953e+00  -7.243
## ticker~XDB    -5.589e+01  7.555e+00  -7.397

```

```
## ticker^XDE      -5.614e+01  7.243e+00 -7.752
## ticker^XDN      -5.635e+01  7.431e+00 -7.583
## tickerAAPL       1.826e+00  6.257e+00  0.292
## tickerAMD        5.354e-02  6.218e+00  0.009
## tickerAMZN      -7.529e-01  6.235e+00 -0.121
## tickerGOOGL      7.206e+00  6.254e+00  1.152
## tickerJNJ       2.158e+01  7.097e+00  3.041
## tickerKO        1.593e+01  6.949e+00  2.292
## tickerMETA       6.853e+00  6.242e+00  1.098
## tickerMSFT       6.434e+00  6.301e+00  1.021
## tickerNFLX      -5.958e+00  6.238e+00 -0.955
## tickerNVDA      -5.767e-01  6.225e+00 -0.093
## tickerPG        1.585e+01  6.818e+00  2.324
```

```
##
## Correlation matrix not shown by default, as p = 28 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
r2(rmr_absolute)
```

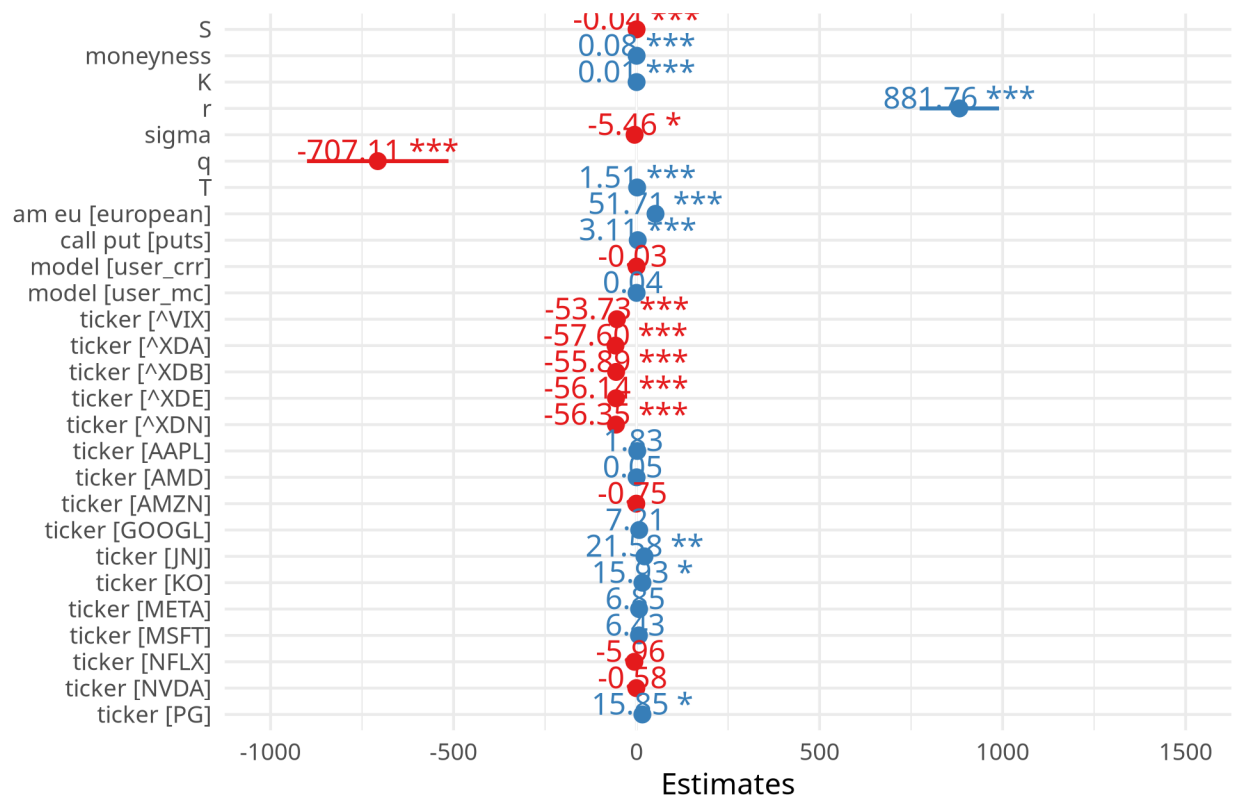
```
## # R2 for Mixed Models
##
##   Conditional R2: 0.035
##   Marginal R2: 0.018
```

```
absolute_bias_plot <- plot_model(rmr_absolute, type = "est", show.values = TRUE, value.offset = .4) +
  theme_minimal() +
  labs(title = "Systematic Biases in Option Pricing Models")
```

```
#ggsave("~/Praxe/bakalarka/figures/absolute_bias.png", absolute_bias_plot, width = 6, height = 4)
```

```
absolute_bias_plot
```

Systematic Biases in Option Pricing Models

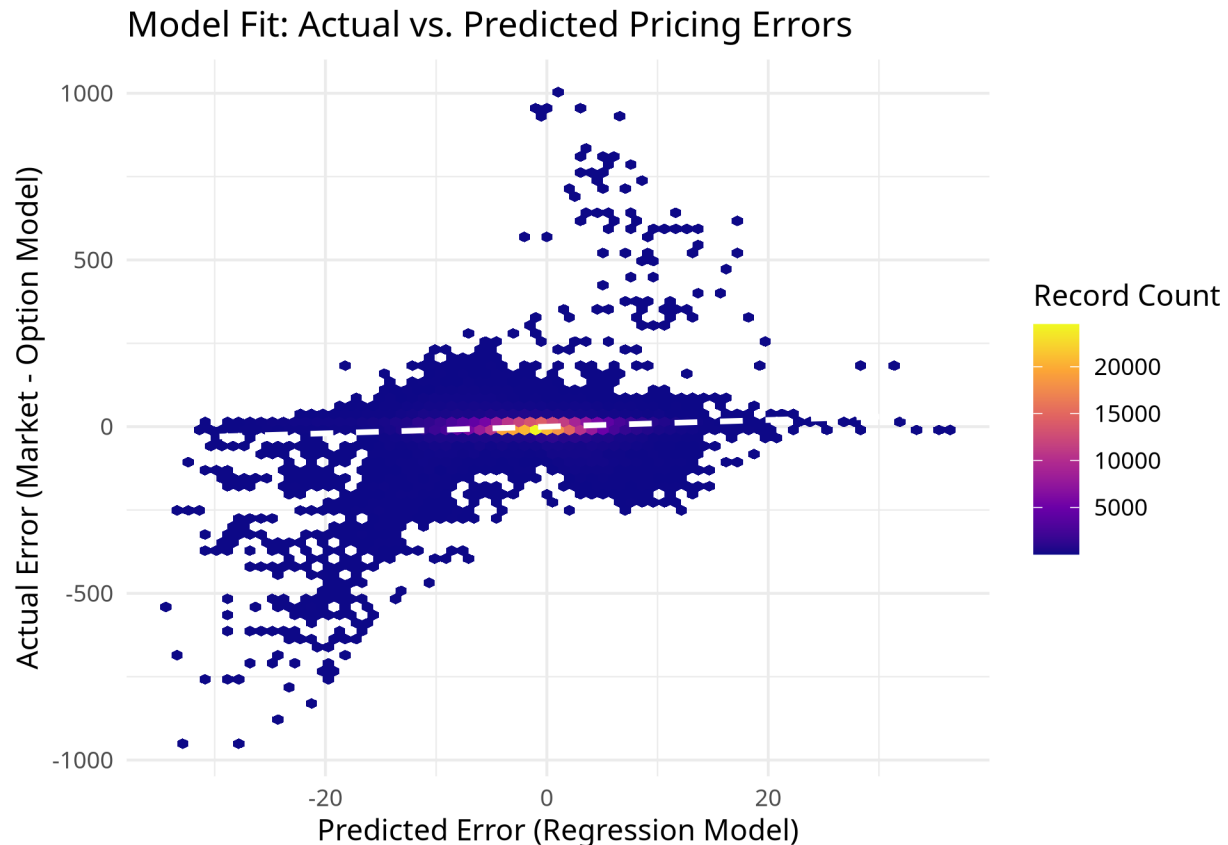


```
train_data_abs$predicted_error <- predict(rmr_absolute)

absolute_hexes <- ggplot(train_data_abs, aes(x = predicted_error, y = pricing_error)) +
  geom_hex(bins = 70) +
  scale_fill_viridis_c(option = "plasma") + # Cool modern color palette
  geom_abline(slope = 1, intercept = 0, color = "white", linetype = "dashed", size = 1) +
  theme_minimal() +
  labs(
    title = "Model Fit: Actual vs. Predicted Pricing Errors",
    x = "Predicted Error (Regression Model)",
    y = "Actual Error (Market - Option Model)",
    fill = "Record Count"
  )

#ggsave("~/Praxe/bakalarka/figures/absolute_hexes.png", absolute_hexes, width = 6, height = 4)

absolute_hexes
```



Relativní

```
df_relative <- df_doctored %>%
  pivot_longer(cols = c(user_bs, user_crr, user_mc),
               names_to = "model",
               values_to = "model_price") %>%
  #filter(model == "user_crr") %>%
  mutate(pricing_error_test = (true_price - model_price) / true_price) %>%
  #mutate(pricing_error = 2 * (true_price - model_price) / (true_price + model_price)) %>%
  mutate(pricing_error = (true_price - model_price) / S) %>%
  filter(is.finite(pricing_error)) %>%
  mutate(snapshot_date = as.Date(expiration) - round(T * 365)) %>%
  mutate(snapshot_month = format(snapshot_date, "%m"))

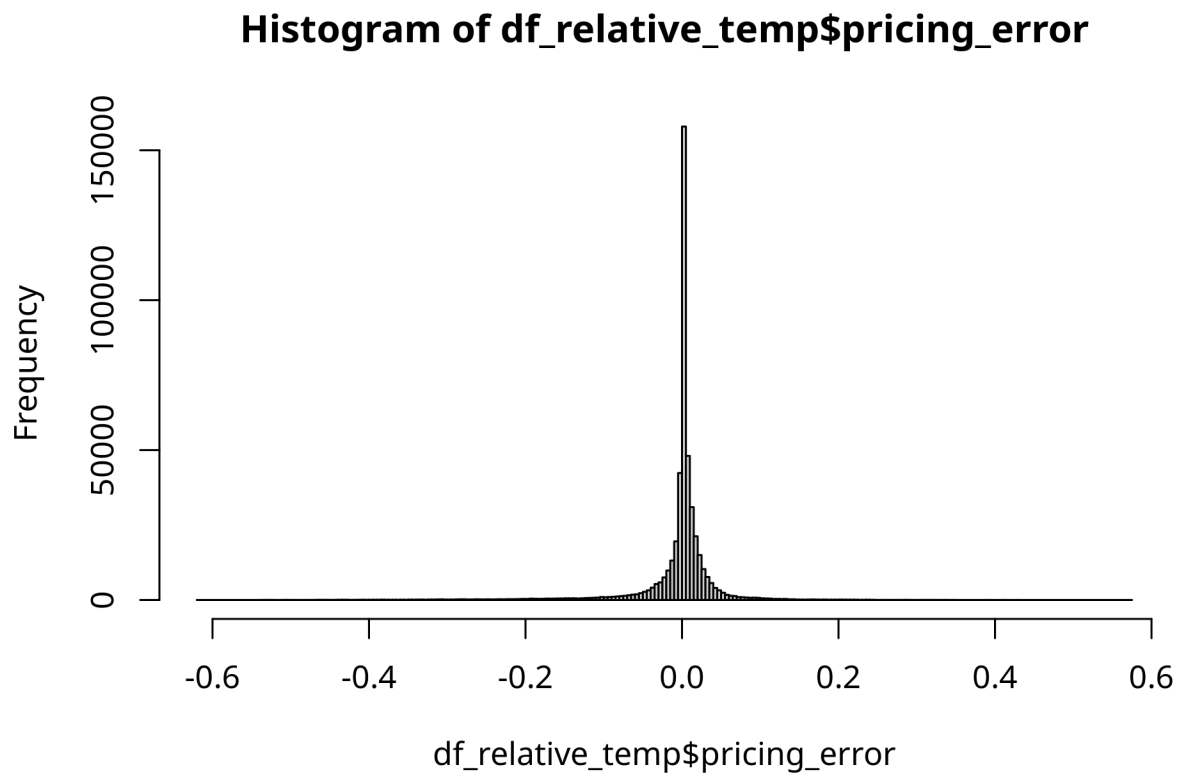
df_relative_temp <- df_relative %>% filter(am_eu == "american" & call_put == "calls")
train_indexes <- sample(1:nrow(df_relative_temp), size = 0.8 * nrow(df_relative_temp))

train_data_rel <- df_relative_temp[train_indexes, ]
test_data_rel <- df_relative_temp[-train_indexes, ]

png("~/Praxe/bakalarka/figures/hist_relative.png", width = 6, height = 4, units = "in", res = 300)
hist(df_relative_temp$pricing_error, breaks = 200)
dev.off()

## png
## 2
```

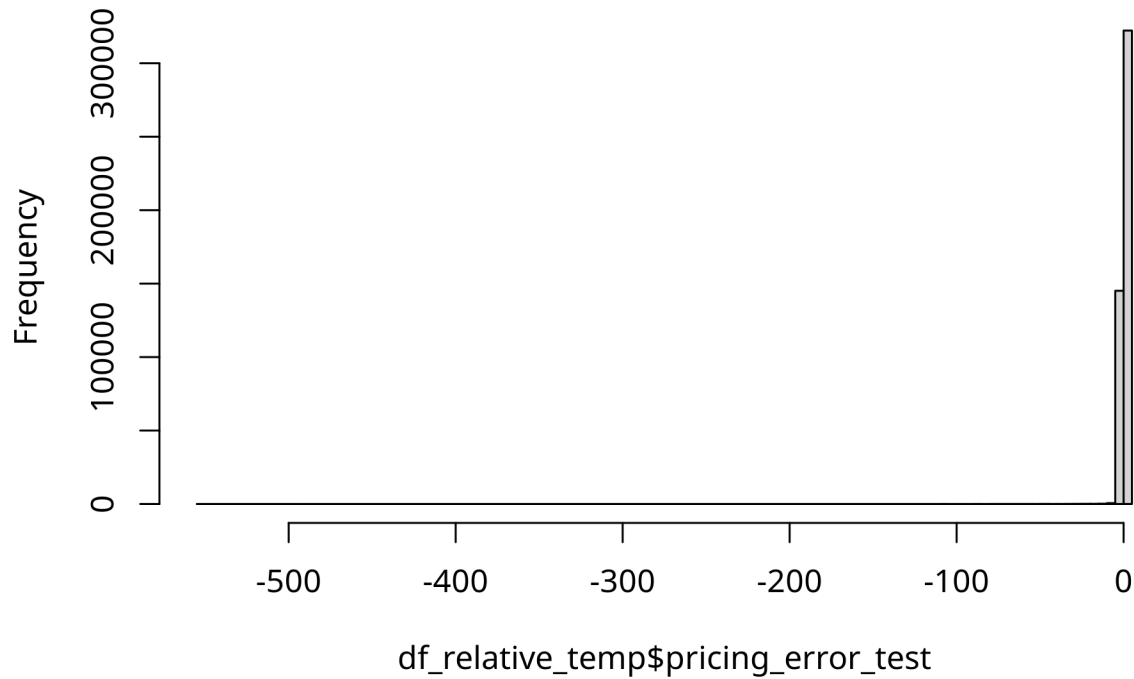
```
hist(df_relative_temp$pricing_error, breaks = 200)
```



Distribuce je bimodální: pík u 0 odpovídá přesnému ocenění, pík u 2 odpovídá opcím s odhadovanou cenou blízkou 0. Regresní model tuto strukturu zohledňuje.

```
hist(df_relative_temp$pricing_error_test, breaks = 100)
```

Histogram of df_relative_temp\$pricing_error_test



```
train_data_rel$T2 <- train_data_rel$T^2

subset_args <- c(
  "S",
  "K",
  #"r",
  "sigma",
  #"q",
  "T",
  "T2",
  #"am_eu",
  #"call_put",
  #"model",
  "ticker",
  "snapshot_month"
)
interactions <- c(
  #"q:am_eu",
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))
#T_crr <- lm(pricing_error ~ T + as.factor(snapshot_month), data = train_data)
regression_relative <- lm(formula, data = train_data_rel)

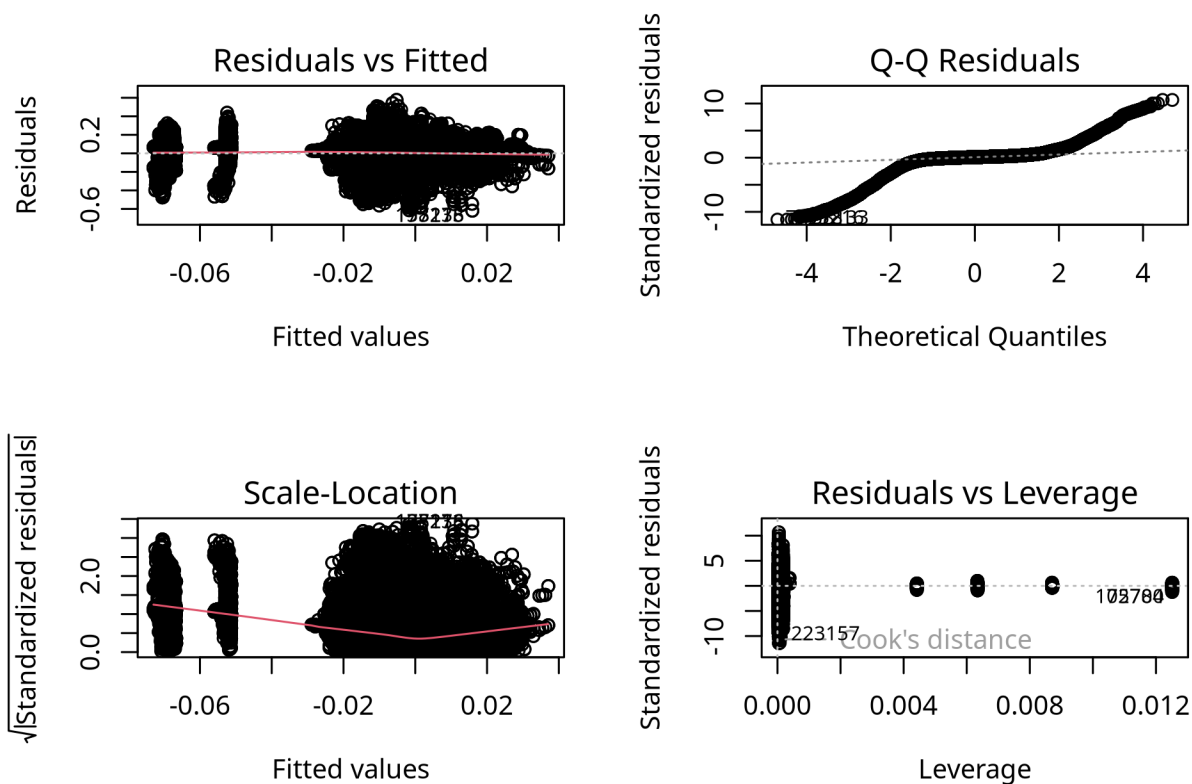
summary(regression_relative)
```

```
##
## Call:
## lm(formula = formula, data = train_data_rel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.61961 -0.00407  0.00549  0.01396  0.57767
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.481e-02  4.899e-03   5.063 4.12e-07 ***
## S             9.588e-06  1.865e-06   5.139 2.76e-07 ***
## K            -1.652e-05  3.940e-07 -41.926 < 2e-16 ***
## sigma        -5.685e-02  3.429e-03 -16.581 < 2e-16 ***
## T            -5.366e-03  3.864e-04 -13.888 < 2e-16 ***
## T2            4.085e-03  1.615e-04  25.287 < 2e-16 ***
## ticker~VIX    -1.514e-02  6.857e-03  -2.208 0.02724 *
## ticker~XDA    -3.370e-02  7.725e-03  -4.363 1.28e-05 ***
## ticker~XDB    -4.759e-03  6.875e-03  -0.692 0.48878
## ticker~XDE    -1.737e-02  5.914e-03  -2.936 0.00332 **
## ticker~XDN    -8.853e-03  6.458e-03  -1.371 0.17039
## tickerAAPL    -1.254e-02  4.568e-03  -2.746 0.00603 **
## tickerAMD      2.852e-03  4.944e-03   0.577 0.56402
## tickerAMZN    -3.442e-03  4.686e-03  -0.734 0.46271
## tickerGOOGL    1.565e-03  4.475e-03   0.350 0.72657
## tickerJNJ     -1.844e-02  4.576e-03  -4.030 5.59e-05 ***
## tickerKO      -1.427e-02  4.861e-03  -2.936 0.00333 **
## tickerMETA    -4.635e-03  4.008e-03  -1.157 0.24740
## tickerMSFT    -1.191e-02  4.267e-03  -2.791 0.00526 **
## tickerNFLX    -1.598e-02  4.936e-03  -3.237 0.00121 **
## tickerNVDA     3.868e-03  4.841e-03   0.799 0.42429
## tickerPG      -1.100e-02  4.739e-03  -2.320 0.02032 *
## tickerTSLA    -3.782e-04  4.631e-03  -0.082 0.93491
## snapshot_month02 2.101e-05  2.552e-04   0.082 0.93440
## snapshot_month03 2.334e-03  2.772e-04   8.419 < 2e-16 ***
## snapshot_month05 4.106e-03  3.321e-04  12.365 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05419 on 375336 degrees of freedom
## Multiple R-squared:  0.05632,    Adjusted R-squared:  0.05626
## F-statistic: 896 on 25 and 375336 DF,  p-value: < 2.2e-16

png("~/Praxe/bakalarka/figures/relative_basic.png", width = 6, height = 4, units = "in", res = 300)
par(mfrow = c(2, 2))
plot(regression_relative)
dev.off()

## png
## 2

par(mfrow = c(2, 2))
plot(regression_relative)
```



```
par(mfrow = c(1, 1))
```

Předpoklady

Kontrola homoscedasticity:

```
bptest(regression_relative)
```

```
##
## studentized Breusch-Pagan test
##
## data: regression_relative
## BP = 18631, df = 25, p-value < 2.2e-16
```

Dle výsledků Breuschova-Paganova testu ($p < 0.05$) zamítáme null hypotézu → je přítomná heteroskedasticita.

Kontrola autokorelace:

```
dwtest(regression_relative)
```

```
##
## Durbin-Watson test
##
## data: regression_relative
## DW = 2.0014, p-value = 0.6702
## alternative hypothesis: true autocorrelation is greater than 0
```

Dle výsledku Durbinova-Watsonova testu je přítomná autokorelace. Opce ze stejného tickeru jsou navzájem závislé.

Použijeme cluster-robust standardní chyby (cluster na ticker), které opravují heteroskedasticitu i autokorelaci zároveň:

Kontrola multikolinearity:

```
vif(regression_relative, type = "predictor")

## GVIFs computed for predictors

##              GVIF Df GVIF^(1/(2*Df)) Interacts With
## S              255.232673  1      15.976003         --
## K              10.911433  1       3.303246         --
## sigma          62.725842  1       7.919965         --
## T              9.964841  1       3.156714         --
## T2             9.748495  1       3.122258         --
## ticker        14276.305338 17       1.324935         --
## snapshot_month  2.911806  3       1.194979         --
##              Other Predictors
## S              K, sigma, T, T2, ticker, snapshot_month
## K              S, sigma, T, T2, ticker, snapshot_month
## sigma          S, K, T, T2, ticker, snapshot_month
## T              S, K, sigma, T2, ticker, snapshot_month
## T2             S, K, sigma, T, ticker, snapshot_month
## ticker         S, K, sigma, T, T2, snapshot_month
## snapshot_month S, K, sigma, T, T2, ticker
```

LMER model - základní

```
subset_args <- c(
  "S",
  "moneyness",
  "K",
  "T",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model",
  "ticker"
)
interactions <- c(
  "q:am_eu",
  #"(1 | option_id)",
  "(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)
```

```
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
## Warning: Some predictor variables are on very different scales: consider rescaling.
```

```
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
summary(rmr_relative)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: pricing_error ~ S + moneyness + K + r + sigma + q + T + am_eu +
##      call_put + model + ticker + q:am_eu + (1 | ticker) + (1 |
##      snapshot_month)
##      Data: train_data_rel
##
## REML criterion at convergence: -1131942
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.8145  -0.1034   0.1061   0.2862  10.6800
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
##      ticker              (Intercept)  1.482e-04  0.0121754
##      snapshot_month (Intercept)  6.883e-07  0.0008296
##      Residual                2.867e-03  0.0535455
## Number of obs: 375362, groups:  ticker, 18; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  -6.722e-02  1.273e-02  -5.279
## S              1.379e-05  1.819e-06   7.582
## moneyness     -1.111e-05  1.086e-05  -1.023
## K             -2.249e-05  4.027e-07 -55.852
## r              2.730e+00  8.555e-02  31.909
## sigma        -5.975e-02  3.319e-03 -18.006
## q            -1.210e+00  1.534e-01  -7.890
## T              4.456e-03  1.260e-04  35.373
## am_eueuropean  3.599e-03  1.779e-02   0.202
## call_putputs  -1.658e-02  1.778e-04 -93.250
## modeluser_crr -4.731e-05  2.140e-04  -0.221
## modeluser_mc   4.068e-05  2.141e-04   0.190
## ticker^VIX     -1.793e-02  1.844e-02  -0.972
## ticker^XDA     -4.762e-02  1.882e-02  -2.530
## ticker^XDB     -2.301e-03  1.850e-02  -0.124
## ticker^XDE     -2.336e-02  1.817e-02  -1.286
## ticker^XDN     -1.993e-02  1.835e-02  -1.086
## tickerAAPL     -8.337e-03  1.726e-02  -0.483
## tickerAMD       2.598e-03  1.723e-02   0.151
## tickerAMZN     -4.290e-03  1.724e-02  -0.249
## tickerGOOGL     5.922e-03  1.726e-02   0.343
## tickerJNJ       2.075e-02  1.803e-02   1.151
## tickerKO        2.111e-02  1.789e-02   1.180
## tickerMETA     -8.708e-04  1.725e-02  -0.050
## tickerMSFT     -2.465e-03  1.730e-02  -0.142
## tickerNFLX     -1.786e-02  1.724e-02  -1.036
## tickerNVDA      3.407e-03  1.723e-02   0.198
## tickerPG        2.077e-02  1.777e-02   1.169
##
```

```

## Correlation matrix not shown by default, as p = 28 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)         if you need it

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

## Random effect variances not available. Returned R2 does not account for random effects.

## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.079

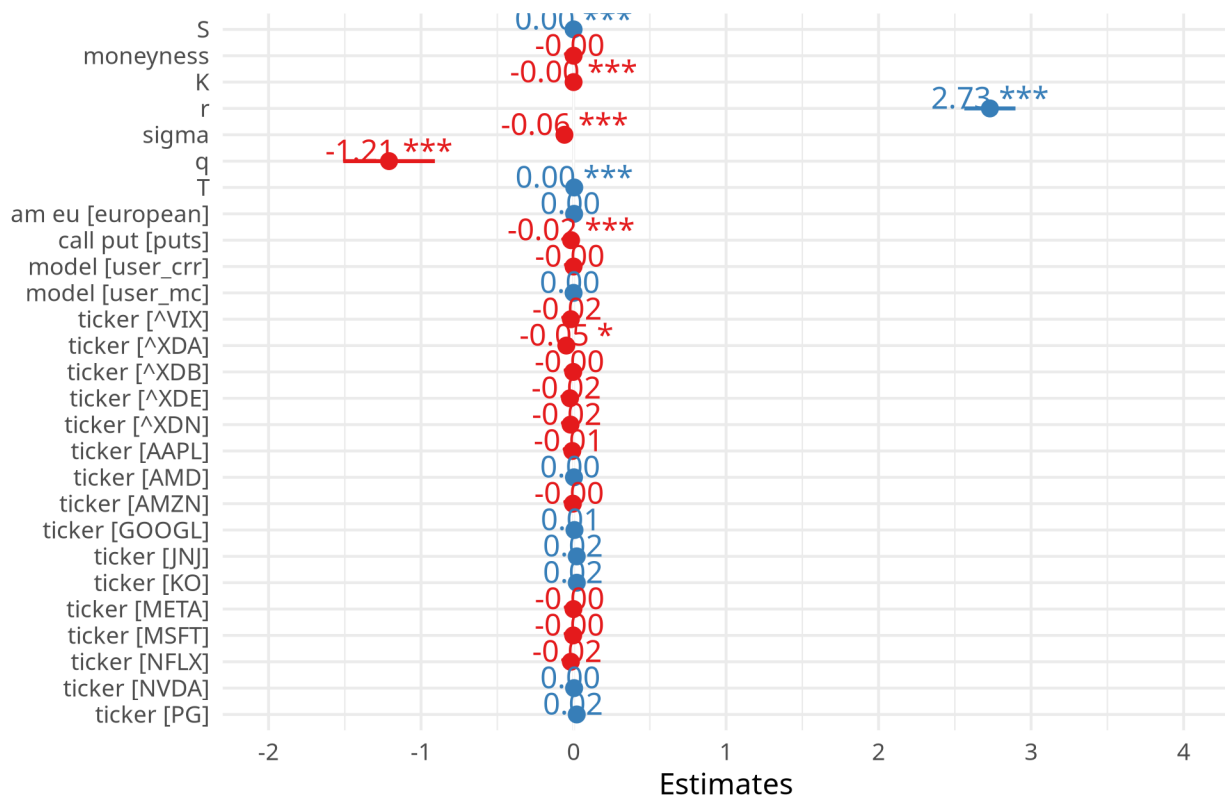
relative_bias_plot <- plot_model(rmr_relative, type = "est", show.values = TRUE, value.offset = .4) +
  theme_minimal() +
  labs(title = "Systematic Biases in Option Pricing Models")

#ggsave("~/Praxe/bakalarka/figures/relative_bias.png", relative_bias_plot, width = 6, height = 4)

relative_bias_plot

```

Systematic Biases in Option Pricing Models



```
train_data_rel$predicted_error <- predict(rmr_relative)

relative_hexes <- ggplot(train_data_rel, aes(x = predicted_error, y = pricing_error)) +
  geom_hex(bins = 70) +
  scale_fill_viridis_c(option = "plasma") + # Cool modern color palette
  geom_abline(slope = 1, intercept = 0, color = "white", linetype = "dashed", size = 1) +
  theme_minimal() +
  labs(
    title = "Model Fit: Actual vs. Predicted Pricing Errors",
    x = "Predicted Error (Regression Model)",
    y = "Actual Error (Market - Option Model)",
    fill = "Record Count"
  )

#ggsave("~/Praxe/bakalarka/figures/relative_predict.png", relative_hexes, width = 6, height = 4)

relative_hexes
```

Model Fit: Actual vs. Predicted Pricing Errors

